

An Iterative Closed-Loop Generative Pipeline for De Novo and Transfer Learning-Guided Small-Molecule Inhibitor Discovery

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September 10, 2026

Abstract

Abstract

Target 200–250 words. Use this six-sentence skeleton and expand each into 1–2 sentences:

1. **Problem.** Chemical space is enormous; experimental screening is slow and expensive.
2. **Gap.** Generative models produce molecules, but iterative target-specific optimisation and honest validation remain open.
3. **What we built.** A closed loop: REINVENT4 generation → Boltz-2 structure-based scoring → gated top- K selection → transfer-learning fine-tune.
4. **How we validated it.** De novo rediscovery: the known-actives set is withheld from the generator and used only to measure convergence via Tanimoto similarity.
5. **Headline result.** Numbers. Actual numbers. $\bar{T}_t$ rose from $\underline{x}$ to $\underline{y}$ over $\underline{n}$ iterations for target; $\underline{n}$ rediscovery hits at $\theta_{\text{hit}} = 0.85$.
6. **Why it matters.** Generalisation across two binding-site architectures (p38 MAPK14, CDK2) plus an interpretable chemical narrative of what the model learns to favour.

Do not put citations in the abstract. Do not use the word “novel”.

[TODO: abstract]

Keywords: generative molecular design; de novo drug design; transfer learning; small-molecule inhibitor discovery; protein–ligand interaction modeling; structure-based scoring.

1 Introduction

How the Introduction is structured, and why

An introduction is an argument, not a summary of everything you read. The argument has a fixed shape and it must end where your paper begins:

Drug discovery needs better search → generative models search well but can't tell good from bad → structure-based affinity models can tell good from bad but don't search → closing the loop between them is the obvious move, and here is what is still unresolved about doing it → our contribution.

Each subsection below is one move in that argument. Write each as 2–4 paragraphs. Delete the `\subsection` headings before submission if the venue prefers a continuous introduction — the headings are scaffolding for you, not necessarily for the reader.

Total citation budget: roughly 35–55 references for the whole introduction. If a subsection below has fewer than 4 citations,

you have not done the literature review for it yet.

1.1 The scale of the search problem

Write §1.1: 2 paragraphs

Establish the stakes with concrete numbers, not adjectives.

- Size of drug-like chemical space (the 10^{23} – 10^{60} estimates — cite the actual source, do not cite a blog post).
- Cost and timeline of bringing one compound to market; attrition rates by phase.
- What high-throughput screening can and cannot cover, and why "more screening" does not close the gap.

Literature to find: Polishchuk et al. on GDB-17 / enumerable chemical space; a recent (2020+) R&D cost-and-attrition analysis; a review of HTS hit rates. **Search terms:** "size of chemical space drug-like", "pharmaceutical R&D productivity attrition", "high-throughput screening hit rate review".

[TODO: write §1.1]

1.2 Generative models for molecular design

Write §1.2: 3–4 paragraphs

This is your main methods-lineage review. Organise it by *representation and objective*, not chronologically by year.

- **Representations:** SMILES-based RNNs/transformers, graph-based generators, 3D/structure-conditioned generators. One paragraph each is plenty; you only need enough to justify why a SMILES-based prior is a reasonable choice here.
- **Optimisation strategies:** reinforcement learning (REINVENT lineage [1–3]), transfer learning / fine-tuning on curated corpora [4], genetic algorithms, Bayesian optimisation, and active learning.
- **The honest limitation:** generated molecules are only as good as the reward. State this plainly — it is the hinge that sets up §1.3.
- **Benchmarking:** GuacaMol, MOSES, and the critique that benchmark-optimal molecules are frequently not chemically sensible. This motivates our §2.9 chemical analysis layer.

Search terms: "de novo molecular generation review 2024", "SMILES recurrent neural network molecule generation", "GuacaMol MOSES benchmark generative chemistry", "reward hacking molecular generative model".

[TODO: write §1.2]

1.3 Structure-based scoring and binding affinity prediction

Write §1.3: 3 paragraphs

Build the case for why Boltz-2 is the scoring function, and be fair about the alternatives you did not use.

- **Classical docking** (AutoDock Vina, Glide, Gold): fast, but scoring-function accuracy for affinity ranking is famously weak. Find a benchmark paper that quantifies this rather than asserting it.
- **ML scoring functions and FEP:** where each sits on the accuracy/cost curve. FEP is accurate and far too slow for 5,000 molecules per iteration — say so with a number.
- **Co-folding foundation models:** AlphaFold2/3 [5, 6], Boltz-1 [7], Boltz-2 [8]. Emphasise what makes Boltz-2 usable here specifically: it emits a binder probability *and* a regression affinity estimate in one call, at a cost that fits an iterative loop.
- **Caveats you must state:** predicted affinity is not measured affinity; performance depends on the quality of the input structure; the model may be susceptible to systematic exploitation by a generator optimising against it.

Search terms: "docking scoring function affinity ranking benchmark", "machine learning

scoring function binding affinity review", "Boltz-2 binding affinity", "co-folding model protein ligand".

[TODO: write §1.3]

1.4 Closed-loop and active-learning pipelines

Write §1.4: 2–3 paragraphs

This is the subsection that positions *our* contribution, so it needs the most careful reading. Find prior work that couples a generator to a structure-based oracle in a loop and describe what each did and where it stopped.

- Docking-in-the-loop generative pipelines; "generate–dock–retrain" systems; DiffDock/AlphaFold-scored generation.
- Self-driving / autonomous lab loops, for contrast: they close the loop on *experimental* measurements, we close it on predicted ones. Be explicit that this is a weaker but far cheaper loop.
- **The gap you are filling.** State it in one sentence you could defend in a talk. Candidate framings: (i) most loops report enrichment of their own scoring function, which is circular; we validate against a withheld set of experimentally measured actives; (ii) most report convergence metrics without characterising *what* the chemistry did; we track scaffolds, functional groups and synthesisability alongside.

Search terms: "closed loop generative docking active learning drug design", "iterative fine-tuning generative model docking score", "self-driving laboratory molecular design".

[TODO: write §1.4]

1.5 Target selection: p38 MAPK14 and CDK2

Write §1.5: 2 paragraphs, one per target

Justify both targets on *three* axes each: biology, data availability, and structural suitability for the scoring model.

- **p38 MAPK14:** serine/threonine kinase in inflammatory signalling; decades of medicinal chemistry; deep, well-defined, hydrophobic ATP site; a large and chemotype-diverse ChEMBL inhibitor set; heavily used as a benchmark in prior computational papers, which makes our results comparable.
- **CDK2:** cell-cycle kinase; different pocket character and a different (smaller) inhibitor landscape; serves as the generalisation test.
- **Say what this pair does and does not test.** Two kinases test transferability across binding-site architecture but *not* beyond the kinase chemotype. Flag this honestly here; return to it in Section 4 as a limitation and future direction.

Literature to find: 2–3 reviews of p38 inhibitor development (including why clinical programs failed — selectivity/toxicity — which is good framing); 2–3 on CDK2 inhibitors; the PDB entries you actually used, cited by their primary papers.

[TODO: write §1.5]

1.6 Rediscovery as validation

Write §1.6: 1–2 paragraphs

Explain and defend the experimental design, because a sceptical reviewer will push on exactly this.

- The known-actives set is ground truth and is *never* shown to the generator — not to initialise it, not in the fine-tuning corpus. The only place it touches the pipeline is threshold calibration (Section 2.5), and even there it only sets a scalar.
- Why this is deliberately conservative: a pipeline that recovers known actives without prior structural knowledge of them is evidence that the scoring and optimisation components track real structure–activity relationships rather than scoring artefacts.

- Anticipate the objection: Boltz-2 was trained on public structural and affinity data that may include these very compounds. Address it directly. Suggested framing: this is a limitation of *any* pretrained oracle; it bounds our claim to "the loop navigates toward chemistry the oracle recognises as active", and we quantify how far that goes rather than assuming it.

[TODO: write §1.6]

1.7 Contributions

Keep this to 4–5 bullets, each a claim you can point to a figure for

Every bullet should map onto a results subsection. If a bullet has no corresponding figure or table, either cut it or go make the figure.

In this work we:

- present a closed-loop pipeline coupling REINVENT4 generation with Boltz-2 structure-based scoring through a gated selection function and iterative transfer learning; [TODO: confirm phrasing]
- calibrate the binder-probability gate τ empirically against the full known-actives benchmark for each target, rather than assuming a threshold; [TODO:]
- validate the loop as a de novo rediscovery benchmark against [n](#) withheld experimentally validated inhibitors across two kinase targets; [TODO:]
- characterise the chemical trajectory of optimisation — scaffold evolution, functional-group enrichment, drug-likeness and synthetic accessibility — across iterations; [TODO:]
- release all configurations, SLURM scripts and analysis notebooks [TODO: repository URL and license].

Figure 1 — pipeline schematic (make this early, it anchors everything)

A single-column block diagram of the closed loop: REINVENT4 $G_t \rightarrow$ Boltz-2 scoring $\rightarrow$ gating function $\rightarrow$ top- $K \rightarrow$ transfer learning $\rightarrow G_{t+1}$, with the known-actives set drawn *off* the main path (dashed) feeding only calibration and the convergence check. **Source:** adapt the flowchart already in the research plan. **Tool:** TikZ, or draw.io exported to PDF. Do not use a raster screenshot.

2 Methods

How to use the Methods section

Everything in this section that is *shared infrastructure* — the loop structure, the notation, the scoring maths, the metric definitions — is already written for you. Do not rewrite it; you would only introduce inconsistencies between sections written by different people.

What you *do* write is inside the orange boxes: the decisions you made, the numbers you chose, the things that went wrong and what you did about them. Write those in past tense, first person plural ("we set", "we observed"), and be specific. "We used a GPU" is not a method; "each Boltz-2 call ran on one NVIDIA A100 with `-diffusion_samples 1` and `-sampling_steps 200`" is.

Use the notation macros. `\pbind`, `\Tbar`, `\TopK`, etc. are defined in the preamble. If you hand-type p_{bind} somewhere it will render differently from everywhere else and a reader will notice.

Algorithm 1 Closed-loop generative optimisation for one target

Require: pretrained prior G_1 ; target structure S ; gate τ ; batch size N ; corpus size K ; hit threshold θ_{hit} ; held-out actives $\mathcal{A}$

```
1: for  $t = 1, 2, \dots, T_{\text{max}}$  do
2:    $\mathcal{B}_t \leftarrow \{m_1, \dots, m_N\} \sim G_t$  ▷ sample SMILES
3:    $\mathcal{V}_t \leftarrow \text{dedup}(\text{sanitise}(\mathcal{B}_t))$  ▷ canonicalise, valence check, drop duplicates
4:   for all  $m \in \mathcal{V}_t$  do
5:      $(p_{\text{bind}}(m), \hat{y}(m)) \leftarrow \text{BOLTZ-2}(S, m)$ 
6:   end for
7:    $\mathcal{P}_t \leftarrow \{m \in \mathcal{V}_t : p_{\text{bind}}(m) \geq \tau\}$ 
8:    $\mathcal{K}_t \leftarrow \arg \text{top-}K_{m \in \mathcal{P}_t} S(m)$  ▷ Equation (7)
9:   compute  $\bar{T}_t, T_t^*$  against  $\mathcal{A}$  ▷ Equations (11) and (12)
10:  if  $T_t^* \geq \theta_{\text{hit}}$  then
11:    record rediscovery hit at iteration  $t$ 
12:  end if
13:   $G_{t+1} \leftarrow \text{TRANSFERLEARN}(G_t, \mathcal{K}_t)$ 
14: end for
```

Table 1: Known-actives benchmark sets and prepared target structures.

Target	UniProt	PDB ID	Resolution (Å)	$ \mathcal{A} $	Median pIC ₅₀
p38 MAPK14	Q16539	PDB	x.xx	3,860	x.xx
CDK2	P24941	PDB	x.xx	1,402	x.xx

2.1 Overview of the closed loop

At iteration t , a generative model G_t samples a batch of N molecules. Each valid molecule is scored against the prepared target structure by Boltz-2, which returns a binder probability and a regression-based potency estimate. A gated scoring function discards molecules below a binder-probability threshold τ and ranks the remainder by predicted potency; the top K molecules form a training corpus used to fine-tune G_t by transfer learning, producing G_{t+1} . Progress is measured against a held-out set $\mathcal{A}$ of experimentally validated inhibitors that is never exposed to the generator. Algorithm 1 states the procedure.

2.2 Datasets and target preparation

2.2.1 Known-actives benchmark sets

For each target we assembled a set $\mathcal{A}$ of experimentally validated small-molecule inhibitors with measured IC₅₀ values, retrieved from ChEMBL [9]. Records were filtered to IC₅₀ endpoints, deduplicated, and canonicalised as SMILES. Measured potencies are expressed throughout as

$$\text{pIC}_{50} = -\log_{10}(\text{IC}_{50} [\text{M}]) . \quad (1)$$

Set sizes after curation are given in Table 1. This set is used exclusively (i) to calibrate the gate τ (Section 2.5) and (ii) as the convergence reference (Section 2.8). It is never used to initialise, condition, or fine-tune the generative model.

Fill Table 1 and write 1 paragraph on curation

The two $|A|$ counts above are the manifest row counts actually docked (`threshold/<TARGET>/manifest.csv`). Verify them against your own `prepare_inputs.py` output before submission and update if they differ. Then write a short paragraph covering:

- the exact ChEMBL query / export settings and the download date;
- what you filtered out and why (assay type, censored values such as $>$ or $<$, salts/mixtures, missing SMILES, un-parseable SMILES);
- how many records went in and how many survived — give both numbers;
- whether duplicate measurements of the same compound were averaged or the highest-confidence record kept.

2.2.2 Protein structure preparation

Write this: 1 paragraph per target

State for each target: the PDB entry chosen and why (resolution, co-crystallised ligand, conformational state — DFG-in/DFG-out matters for p38); crystallographic waters and heteroatoms removed; hydrogens added; protonation state assignment and at what pH; how the binding site was defined for the Boltz-2 input; and the exact protein sequence used in the YAML (state whether it is the full construct or a truncation, and give residue ranges). If you used a predicted structure anywhere, say so and note the caveat that scoring quality is contingent on structural accuracy [6].

[TODO: structure preparation paragraphs]

2.2.3 Molecular representation

All generated and reference molecules are canonicalised as SMILES using RDKit [10]. For similarity computations, each molecule m is mapped to a Morgan (ECFP4-equivalent) fingerprint [11]

$$\varphi(m) \in \{0, 1\}^{2048}, \quad \text{radius } r = 2, \text{ fold size } 2048, \quad (2)$$

and the Tanimoto coefficient between two molecules is

$$T(\varphi(m), \varphi(m')) = \frac{|\varphi(m) \wedge \varphi(m')|}{|\varphi(m) \vee \varphi(m')|} \in [0, 1]. \quad (3)$$

2.3 Molecular generation

Generation uses REINVENT4 [3] in sampling mode (`run_type = "sampling"`), which draws from the model without modifying its weights. At $t = 1$ the model G_1 is the unmodified pretrained prior `reinvent_pubchem.prior`, trained on PubChem and carrying no knowledge of either target; for $t \geq 2$, G_t is the checkpoint produced by the transfer-learning update of iteration $t - 1$ (Section 2.7). Iteration 1 is therefore the only round sampled from a fully unbiased distribution.

Each iteration samples $N = 5,000$ SMILES per target with `unique_molecules = true` (post-hoc deduplication) and `randomize_smiles = true` (atom-ordering augmentation during sampling). The batch size is a fixed design choice applied uniformly to both targets and is unrelated to the size of either target’s benchmark set. Sampled strings are sanitised and canonicalised with RDKit; molecules failing valence or sanitisation checks are discarded, so $|\mathcal{V}_t| \leq N$.

Write 1 short paragraph + fill Table 2

Report, per target and per iteration: how many of the 5,000 sampled strings survived canonicalisation and deduplication, and whether that fraction changed as iterations progressed. A *falling* valid fraction across iterations is worth commenting on — it can indicate the fine-tuned model drifting toward degenerate strings. Also state the software version (`reinvent -version`) and the random seed if you set one.

Table 2: REINVENT4 sampling configuration (identical across targets and iterations).

Parameter	Value	Note
<code>run_type</code>	<code>sampling</code>	weights unchanged
<code>model_file</code>	<code>prior ($t = 1$) / fine-tuned ckpt ($t \geq 2$)</code>	Section 2.7
<code>num_smiles</code>	5,000	fixed by design, both targets
<code>unique_molecules</code>	<code>true</code>	post-hoc deduplication
<code>randomize_smiles</code>	<code>true</code>	atom-order augmentation
<code>device</code>	<code>cuda:0</code>	one NVIDIA A100
REINVENT4 version	x.y.z	

2.4 Structure-based scoring with Boltz-2

Every molecule is evaluated by a single Boltz-2 [8] call taking the target protein sequence and one ligand SMILES, with an explicit `properties: affinity: binder` declaration in the input YAML so that the affinity head is invoked. Each call returns:

- a predicted complex structure and confidence metrics (pTM, ipTM, pLDDT);
- $p_{\text{bind}}(m) \in [0, 1]$, the predicted probability that m is a binder at the site (`affinity_probability_binary`);
- $\hat{y}(m) = \log_{10} \widehat{\text{IC}}_{50}(m)$ with $\widehat{\text{IC}}_{50}$ in μM (`affinity_pred_value`); lower is more potent.

For comparison with experimental values on a common scale we convert

$$\widehat{\text{pIC}}_{50}(m) = 6 - \hat{y}(m), \quad (4)$$

which follows from Equation (1) and the μM unit convention.

Identical inference settings were used for calibration and for scoring generated molecules: `-model boltz2,-diffusion_samples 1,-sampling_steps 200,-diffusion_samples_affinity 5,-affinity_mw_correction`.

Write 1 paragraph

Cover: Boltz-2 version/commit; wall-clock cost per molecule and total GPU-hours consumed per target; how failures were handled (molecules that produced no affinity JSON and were dropped, and how many); whether the calibration array job and the generated-molecule docking job used the same code path (they should — if they differed at all, say how). If you hit the 24 h walltime on the generated-molecule docking run, describe what you did and whether any molecules were re-docked.

2.5 Threshold calibration

Why this subsection exists

τ is the one number in the whole pipeline chosen by looking at the actives. Reviewers will ask whether that leaks information. Your defence is that τ is a scalar derived from a *score distribution*, not from any structural feature of any active, and that it is fixed before iterative optimisation begins. Make sure the text says this explicitly.

Before iterative optimisation, we characterised the Boltz-2 scoring landscape for each target. We evaluated (i) the complete known-actives set $\mathcal{A}$ as a positive control and (ii) a background sample of $n_{bg} = 500\text{--}1,000$ molecules drawn from the unmodified prior with no reward signal.

Two properties were required before committing a target to iterative runs. First, *separation*: the actives must score above the background on both outputs. We quantify this with the rank-biserial form of the Mann–Whitney U statistic, equivalent to the probability that a randomly chosen active out-scores a randomly chosen background molecule,

$$\text{AUC} = \Pr[p_{\text{bind}}(a) > p_{\text{bind}}(b)], \quad a \sim \mathcal{A}, b \sim \text{background}, \quad (5)$$

with $\text{AUC} = 0.5$ indicating no discrimination. Second, *potency correlation*: predicted versus experimental potency across $\mathcal{A}$, reported as Pearson r and Spearman ρ between $\widehat{\text{pIC}}_{50}$ (Equation (4)) and measured pIC_{50} .

The gate τ is then chosen to sit meaningfully above the background median while remaining attainable by a non-trivial fraction of unbiased samples:

$$\tau = \arg \max_{\tau' \in \{0.3, 0.4, 0.5\}} \left\{ \tau' : \Pr_b[p_{\text{bind}}(b) \geq \tau'] \geq \epsilon \right\}, \quad (6)$$

where $\epsilon = 0.0x$ is the minimum background pass rate needed to keep the transfer-learning corpus populated in early iterations. A gate that excludes every unbiased molecule suppresses the learning signal entirely; a gate that admits the majority pollutes the corpus with non-binders.

Fill Table 3 and write 1–2 paragraphs

Report per target: the p_{bind} percentiles (10/25/50/75/90) for both actives and background; the fraction of each population clearing 0.3, 0.4, 0.5; the AUC from Equation (5); Pearson r and Spearman ρ ; and the τ you finally chose *with the sentence of reasoning that led to it*. If the two targets ended up with different τ , say so and justify it — that is a legitimate finding, not an inconsistency to hide. Also fix ϵ in Equation (6) to the value you actually used, or replace the equation with a plain sentence if you selected τ by inspection.

Table 3: Boltz-2 calibration summary. Percentiles are of p_{bind} .

Target	Population	P_{10}	P_{25}	P_{50}	P_{75}	P_{90}	$\Pr[p_{\text{bind}} \geq \tau]$
p38 MAPK14	known actives						
	prior background						
CDK2	known actives						
	prior background						

2.6 Gated scoring function and candidate selection

The optimisation objective is a gated score. For molecule m ,

$$S(m) = \begin{cases} -\hat{y}(m), & p_{\text{bind}}(m) \geq \tau, \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where the negation makes lower predicted IC_{50} (higher potency) correspond to a higher score, consistent with maximisation. Molecules below the gate are excluded from all downstream steps. Among the survivors $\mathcal{P}_t = \{m \in \mathcal{V}_t : p_{\text{bind}}(m) \geq \tau\}$, candidates are ranked by $S(m)$ and the top K retained:

$$\mathcal{K}_t = \arg \text{top-} K_{m \in \mathcal{P}_t} S(m), \quad K \in [200, 500]. \quad (8)$$

When $|\mathcal{P}_t| < K$ the entire passing set is used.

Alternative scoring formulations — describe these only if you ran them

The plan lists two alternatives. If you ran either as an ablation, report it in Section 3.8; if not, mention them in one sentence in Section 4 as untested design choices rather than claiming them.

- **Option A, composite:** $S_A(m) = w_1 p_{\text{bind}}(m) + w_2 (-\hat{y}(m))$ — a soft gradient over both outputs, at the risk of admitting high- p_{bind} non-binders.
- **Option C, hard filter then potency only:** p_{bind} used purely as a binary filter, ranking by $\hat{y}$ alone — a cleaner potency signal that discards the probabilistic gradient which may be informative when few molecules pass.

State your actual τ and K , and why

One short paragraph. Give the final values used for each target. If K was adapted because too few molecules passed the gate in early iterations, describe the rule you used — an unstated adaptive K is a reproducibility hole.

2.7 Transfer-learning update

The corpus $\mathcal{K}_t$ is used to fine-tune G_t in REINVENT4 transfer learning mode (`run_type = "transfer_learning"`), shifting the model’s output distribution toward the chemical features associated with high $S(m)$ and producing the checkpoint G_{t+1} . Formally, fine-tuning minimises the negative log-likelihood of the selected corpus under the model,

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{K}_t|} \sum_{m \in \mathcal{K}_t} \sum_{i=1}^{|m|} \log P_{\theta}(s_i \mid s_{<i}), \quad (9)$$

where $s_1, \dots, s_{|m|}$ are the SMILES tokens of m and θ is initialised from G_t rather than from scratch. Each iteration’s prior is therefore the previous iteration’s fine-tuned checkpoint, which is what makes the loop compounding.

Fill Table 4 and write 1 paragraph

Hyperparameters must be reported — this is the step that determines whether the loop converges or collapses. Also address: were these held fixed across all iterations (they should be, unless an ablation says otherwise)? Did you monitor validation loss or early-stop? Did you observe mode collapse — a sharp drop in unique molecules or unique scaffolds after a fine-tune — and if so at which iteration?

Table 4: Transfer-learning hyperparameters, held fixed across iterations.

Parameter	Value
Corpus size K	
Epochs	
Learning rate	
Batch size	
Optimiser	
Validation split	none / x%
Early stopping	

2.8 Convergence metrics

Convergence is measured by structural similarity between generated molecules and the held-out actives $\mathcal{A}$. For each generated molecule m we define its nearest-neighbour similarity to the benchmark set,

$$\text{sim}_{\max}(m, \mathcal{A}) = \max_{a \in \mathcal{A}} T(\varphi(m), \varphi(a)). \quad (10)$$

Two summary statistics are recorded per iteration. The *mean nearest-neighbour similarity* over the selected set,

$$\bar{T}_t = \frac{1}{|\mathcal{K}_t|} \sum_{m \in \mathcal{K}_t} \text{sim}_{\max}(m, \mathcal{A}), \quad (11)$$

is the primary convergence signal: a monotonically increasing $\bar{T}_t$ indicates the loop is navigating toward known-active chemical space. The *maximum* similarity,

$$T_t^* = \max_{m \in \mathcal{K}_t} \text{sim}_{\max}(m, \mathcal{A}), \quad (12)$$

is used for hit detection: molecule m is declared a *rediscovery hit* when $\text{sim}_{\max}(m, \mathcal{A}) \geq \theta_{\text{hit}}$, with $\theta_{\text{hit}} = 0.85$ for near-identical scaffolds and $\theta_{\text{hit}} = 1.0$ for an exact canonical SMILES match. The first iteration at which $T_t^* \geq \theta_{\text{hit}}$ is recorded as the convergence iteration. As a null reference we also report $\bar{T}_0$, computed identically on an unbiased sample from the prior, so that any rise in $\bar{T}_t$ is read against the level expected by chance.

Because the convergence iteration is not known in advance, no fixed stopping criterion is imposed a priori; $\bar{T}_t$ and T_t^* are monitored per iteration and the run terminates on hit detection or on reaching $T_{\max} = \underline{n}$ iterations.

Two things to compute that the plan does not require but reviewers will want

1. **A random-baseline curve.** Run the metric on K molecules sampled from the untouched prior at every iteration. If $\bar{T}_t$ rises and the baseline is flat, your effect is real. Without it, the rise is uninterpretable.
2. **Error bars.** $\bar{T}_t$ is a mean over K molecules — report the standard error, or better, bootstrap a 95% CI. A convergence curve with no uncertainty band is the most common criticism of this kind of plot.

2.9 Iterative chemical analysis

In parallel with convergence tracking, the selected set $\mathcal{K}_t$ is characterised chemically at each iteration.

Scaffold tracking. Each molecule is reduced to its Bemis–Murcko scaffold [12], $\text{Mur}(m)$, using the RDKit implementation. Scaffold diversity at iteration t is the normalised count of distinct scaffolds,

$$D_t = \frac{|\{\text{Mur}(m) : m \in \mathcal{K}_t\}|}{|\mathcal{K}_t|} \in (0, 1], \quad (13)$$

with $D_t \rightarrow 0$ indicating collapse onto a dominant chemotype. We additionally report the scaffold overlap with the benchmark set, $|\{\text{Mur}(m) : m \in \mathcal{K}_t\} \cap \{\text{Mur}(a) : a \in \mathcal{A}\}|$, and the scaffold transition between consecutive iterations, comparing the most frequent scaffolds of $\mathcal{K}_t$ and $\mathcal{K}_{t+1}$ to identify which ring systems the model retains, discards, or introduces.

Functional-group trajectories. For a set $\mathcal{G}$ of SMARTS patterns covering hydrogen-bond donors and acceptors (hydroxyl, amine, carbonyl, amide), halogens (F, Cl, Br), aromatic systems, and basic nitrogen heterocycles (piperidine, piperazine, morpholine), the frequency of group g at iteration t is

$$f_{g,t} = \frac{1}{|\mathcal{K}_t|} \sum_{m \in \mathcal{K}_t} \mathbb{K}[m \text{ matches } g]. \quad (14)$$

Monotone trends in $f_{g,t}$ are interpreted against the known character of the target binding site. The amine pattern excludes amide nitrogens so that amides are not double-counted.

R-group decomposition. Where a dominant core emerges, molecules matching it are decomposed into core plus substituents at each attachment point, and substituents at position R_j are ranked by mean predicted score,

$$\bar{S}_{R_j=v} = \frac{1}{|\{m : R_j(m) = v\}|} \sum_{m : R_j(m)=v} S(m), \quad (15)$$

yielding a computational structure–activity table.

Drug-likeness and synthesisability. For every molecule in $\mathcal{K}_t$ we compute molecular weight, calculated $\log P$, hydrogen-bond donor and acceptor counts and topological polar surface area; the fraction violating each Lipinski threshold [13] ($\text{MW} > 500$, $c \log P > 5$, $\text{HBD} > 5$, $\text{HBA} > 10$); QED [14]; and the Ertl–Schuffenhauer synthetic accessibility score [15], on a scale from 1 (easy) to 10 (difficult). Together these define a chemical feasibility envelope; systematic drift outside it across iterations is flagged as possible over-optimisation against the scoring function at the expense of chemical viability.

Red-flag screening. Candidates are screened against the RDKit PAINS filter catalogue [16]. A PAINS match is recorded as an annotation for expert review, not as an automatic disqualification.

Descriptor-space projection. A PCA is fitted on the standardised descriptor matrix of $\mathcal{A}$ alone (MW , $c \log P$, HBD , HBA , TPSA , QED) and each iteration’s $\mathcal{K}_t$ is projected into that fixed space. Fitting on the actives only keeps the axes constant across iterations so that successive plots are directly comparable.

Write 1 paragraph on expert structural annotation

Describe your manual review protocol from the plan’s Section 3.6: how many molecules per iteration (5–10), how they were selected (diversity-prioritised, not just top-scoring), who reviewed them, and what was recorded — structural novelty relative to $\mathcal{A}$ and the literature, qualitative synthetic feasibility, plausibility of the predicted binding mode, and red-flag substructures.

Say plainly that this step is qualitative.

2.10 Computational environment and reproducibility

All generation, scoring, and fine-tuning jobs ran on the Anvil cluster (Purdue University / NSF ACCESS) under the SLURM scheduler, using NVIDIA A100 GPUs. REINVENT4 and Boltz-2 were installed in separate conda environments (Python [3.11](#) and [3.10](#) respectively) because of incompatible PyTorch and CUDA dependencies. Chemical analysis was performed in Python with RDKit [\[10\]](#), pandas, scikit-learn, and matplotlib.

Fill in and complete

Report software versions for everything in Table 5, total GPU-hours per target, and the repository URL and commit hash corresponding to the results in this paper. A version table is the cheapest possible reproducibility win — do not skip it.

Table 5: Software versions.

Component	Version	Component	Version
REINVENT4		RDKit	
Boltz-2		scikit-learn	
PyTorch		pandas	
CUDA	12.6	Python	

Data and code availability. [TODO: repository URL, commit hash, DOI/Zenodo archive, license]

3 Results

How to use the Results section — read this before writing anything

This section is deliberately built as a **living tracker first, paper second**. Section [3.1](#) is a status board you update as work lands; Sections [3.3](#) and [3.4](#) contain one short log block per iteration that you fill in the same week the iteration finishes, while you still remember what happened.

When it is time to write the actual prose Results, you will not be starting from a blank page — you will be summarising your own logs. That is the entire point of the structure. Keep the logs even after the prose exists; move them to an appendix or delete them at the very end.

Rules for the prose you eventually write:

- Results state what happened. Interpretation goes in Section [4](#). " $\bar{T}_t$ rose from 0.21 to 0.34" belongs here; "suggesting the model learned the hinge-binding motif" does not.
- Every number in the text must also appear in a table or figure, and vice versa.
- Report the failures. An iteration where $\bar{T}_t$ dropped, a target where calibration showed no separation, a fine-tune that collapsed diversity — these make the paper more credible, not less.

3.1 Pipeline progress tracker

Legend and instructions

Update Tables 6 and 7 as each stage completes. Glyphs: ✓ complete and pushed to main ■ in progress / job running ✗ blocked or failed ○ not started. Add a new row for each iteration as it begins. The "Owner" column is who to ask when something in that row is unclear six weeks from now.

Table 6: Execution status — p38 MAPK14. Owner initials in parentheses.

Iter	Stage set	Gen	Dock	Gate + TL	Chem. analysis	Last update
0	calibration (τ)	—	■ (MB)	—	○	date
1	prior sampling	✓ ()	■ ()	○	○	date
2		○	○	○	○	date
3		○	○	○	○	date

Table 7: Execution status — CDK2. Owner initials in parentheses.

Iter	Stage set	Gen	Dock	Gate + TL	Chem. analysis	Last update
0	calibration (τ)	—	■ (AK)	—	○	date
1	prior sampling	✓ ()	■ ()	○	○	date
2		○	○	○	○	date
3		○	○	○	○	date

Table 8 is the quantitative counterpart: one row per completed iteration per target, and the table that most of the Results prose will be written from.

3.2 Boltz-2 calibration

Write 2–3 paragraphs. This is the first real result in the paper.

Report, per target: whether known actives separated from the prior background on p_{bind} , by how much (Equation (5)), and how predicted potency correlated with measured potency (r , ρ , n). Then state the chosen τ and the fraction of background molecules it admits.

This subsection carries a go/no-go decision, so be direct about it. If a target showed weak or absent separation, say so here and explain what you concluded — structural preparation issue, or a genuine limitation of Boltz-2 for that target. Either is a publishable finding. Burying it is not an option, because a reader can tell from the correlation plot.

[TODO: calibration results]

Figure 2 — calibration (2 panels)

(a) p_{bind} distributions, known actives vs. prior background, overlaid density histograms per target, with vertical lines at $\tau \in \{0.3, 0.4, 0.5\}$. (b) Predicted vs. experimental pIC_{50} scatter, with $y = x$ reference line, Pearson r annotated, and a marginal histogram. **Source:** `threshold/<TARGET>/analyze_calibration.py` already produces `p_bind_distribution.png` and `pIC50_correlation.png`; combine into one two-panel figure with shared styling rather than pasting two separate PNGs. **Watch for:** save as PDF or 300 dpi PNG. The default 150 dpi in the tutorial script is fine for a Colab preview and too low for print.

Table 8: Master results tracker. One row per completed iteration. $|\mathcal{V}_t|$: valid unique molecules of $N = 5,000$ sampled. $|\mathcal{P}_t|$: molecules with $p_{\text{bind}} \geq \tau$. $\bar{S}$: mean gated score over $\mathcal{K}_t$. $\bar{T}_t, T_t^*$: Equations (11) and (12). D_t : scaffold diversity, Equation (13). Hits: molecules with $\text{sim}_{\text{max}} \geq \theta_{\text{hit}} = 0.85$.

Target	t	$ \mathcal{V}_t $	$ \mathcal{P}_t $	K	$\bar{S}$	$\bar{T}_t$	T_t^*	Hits	D_t	$\overline{\text{QED}}$	$\overline{\text{SA}}$
p38 MAPK14	0 ^a	—	—	—	—						
	1										
	2										
	3										
CDK2	0 ^a	—	—	—	—						
	1										
	2										
	3										

^aRow $t = 0$ is the unbiased-prior null reference: the same metrics computed on K molecules sampled from the untouched prior. Every claim of improvement is relative to this row.

3.3 p38 MAPK14

Per-iteration log blocks — fill one in per iteration, the week it finishes

Copy the block below for each new iteration. Keep them terse; they are notes to yourself, and later the raw material for the prose. The "Observations" field is the one that matters most and the one people skip — write two honest sentences about what the chemistry looked like and anything that surprised you.

Iteration 1 — unbiased prior

Log

Date completed
 Model G_1 `reinvent_pubchem.prior` (no fine-tuning)
 Generation job [SLURM job ID](#)
 Docking job [SLURM job ID](#)
 $|\mathcal{V}_1|/N$ / 5,000
 τ used
 $|\mathcal{P}_1|, K$,
 $\bar{T}_1 / T_1^*$ /
 Hits (≥ 0.85)
 Scaffold count / D_1 /
 Shared scaffolds with $\mathcal{A}$
 Mean QED / SA /
 Observations [Two sentences: what did the top- \$K\$ chemistry look like? Anything unexpected?](#)
 Problems / deviations [Timeouts, dropped molecules, anything you did differently from the protocol.](#)
Expected at $t = 1$: zero or near-zero rediscovery hits and few or no shared scaffolds. G_1 has had no transfer learning and no knowledge of either target. This row establishes the floor everything else is measured against — it is a result, not a failure.

Iteration 2

Log		
Date completed		
Model G_2		checkpoint path from iteration 1 fine-tune
Generation job		
Docking job		
$ \mathcal{V}_2 /N$		/ 5,000
$ \mathcal{P}_2 , K$		,
$\bar{T}_2 / T_2^*$		/
$\Delta\bar{T}$ vs. $t = 1$		
Hits (≥ 0.85)		
Scaffold count / D_2		/
New dominant scaffolds		which ring systems appeared that weren't in $\mathcal{K}_1$?
Mean QED / SA		/
Observations		
Problems / deviations		

Iteration 3

Log — copy this block for iterations 4, 5, ...		
Date completed		
Model G_3		
$ \mathcal{V}_3 /N$		/ 5,000
$ \mathcal{P}_3 , K$		,
$\bar{T}_3 / T_3^*$		/
$\Delta\bar{T}$ vs. $t = 2$		
Hits (≥ 0.85)		
Scaffold count / D_3		/
Mean QED / SA		/
Observations		
Problems / deviations		

Then write the p38 prose: 3–4 paragraphs

Walk the reader down the columns of Table 8, in this order:

1. **Throughput and gating.** How many molecules survived validity filtering, how many cleared τ , and how the pass rate moved across iterations. A rising pass rate is itself evidence the fine-tuning is working, independent of Tanimoto.
2. **Score enrichment.** How the distribution of $S(m)$ shifted. Give median and IQR, not just the mean.
3. **Convergence.** $\bar{T}_t$ and T_t^* against the $t = 0$ null. State whether the trend is monotone and where it plateaus.
4. **Hits.** How many, at which iteration, and what they resemble. If there are none, say that clearly and report the closest pairs instead — that is still a quantitative result.

[TODO: p38 results prose]

Figure 3 — convergence curves (the paper's headline figure)

$\bar{T}_t$ (line, with bootstrap 95% CI band) and T_t^* (line, second axis or second panel) versus iteration t , one colour per target on shared axes, plus a flat dashed *prior-baseline* line at the $t = 0$ value and a horizontal dashed line at $\theta_{\text{hit}} = 0.85$. **Why it matters:** this single panel is the paper's central claim. Make it early, even with only two iterations of data — looking at it will tell you whether the loop is working long before the full run finishes.

Figure 4 — score distribution evolution

Violin or ridgeline plot of $S(m)$ over $\mathcal{K}_t$, one violin per iteration, faceted by target. Overlay the median. This is the plan's "progressive enrichment" visualisation.

3.4 CDK2

Same log structure — copy the p38 blocks above

Keep the fields identical to Section 3.3 so the two targets stay directly comparable. Resist the urge to track extra things for one target only; asymmetric records make the cross-target comparison in Section 3.7 much harder to write.

Iteration 1 — unbiased prior

Log

Date completed	
Model G_1	reinvent_pubchem.prior (no fine-tuning)
Generation job	
Docking job	
$ \mathcal{V}_1 /N$	/ 5,000
τ used	
$ \mathcal{P}_1 , K$	,
$\bar{T}_1 / T_1^*$	/
Hits (≥ 0.85)	
Scaffold count / D_1	/
Shared scaffolds with $\mathcal{A}$	
Mean QED / SA	/
Observations	
Problems / deviations	

Iteration 2

Log

Date completed	
Model G_2	
$ \mathcal{V}_2 /N$	/ 5,000
$ \mathcal{P}_2 , K$	,
$\bar{T}_2 / T_2^*$	/
$\Delta \bar{T}$ vs. $t = 1$	
Hits (≥ 0.85)	
Scaffold count / D_2	/
New dominant scaffolds	
Mean QED / SA	/
Observations	
Problems / deviations	

Iteration 3

Log — copy for later iterations

Date completed	
Model G_3	
$ \mathcal{V}_3 /N$	/ 5,000
$ \mathcal{P}_3 , K$	,
$\bar{T}_3 / T_3^*$	/
Hits (≥ 0.85)	
Scaffold count / D_3	/
Mean QED / SA	/
Observations	
Problems / deviations	

Then write the CDK2 prose: 3–4 paragraphs

Same four-part structure as p38. Where CDK2 behaved differently, say so here in factual terms and save the "why" for Section 4. Note that CDK2's benchmark set is roughly a third the size of p38's, which affects $\text{sim}_{\max}$ directly: a smaller $\mathcal{A}$ means fewer chances for any generated molecule to find a close neighbour. Mention this when comparing raw $\bar{T}_t$ values across targets — otherwise the comparison is not apples to apples.

[TODO: CDK2 results prose]

3.5 Chemical evolution across iterations

Write 3–4 paragraphs. This is the interpretability contribution.

Structure it as: scaffolds $\rightarrow$ functional groups $\rightarrow$ feasibility envelope.

- **Scaffold trajectory.** Did D_t (Equation (13)) hold or collapse? Which scaffolds emerged, at which iteration, and do any resemble known pharmacophores for the target? Name the ring systems explicitly — a reader who knows kinase chemistry will care whether an adenine mimetic or a hinge-binding motif appeared.
- **Functional groups.** Which $f_{g,t}$ trends were monotone? Interpret against binding-site character (a deep hydrophobic ATP site should not be enriching hydroxyls indefinitely). Only claim a trend if it holds over ≥ 3 iterations.
- **Feasibility envelope.** Did MW, $c \log P$, QED and SA stay within bounds? Report the fraction of $\mathcal{K}_t$ violating each Lipinski threshold per iteration. If SA rose steadily, that is a real finding about over-optimisation and belongs in the abstract.
- **PAINS.** What fraction of top candidates carried a flag, and which catalogue entries.

[TODO: chemical evolution prose]

Figure 5 — chemical trajectory (multi-panel, one row per metric)

A grid: rows = {scaffold diversity D_t , mean QED, mean SA score, mean MW, mean $c \log P$ }, columns = {p38, CDK2}. Each panel is a line over iteration with a shaded band for the known-actives reference range (e.g. the actives' interquartile range as a horizontal band). The band is what makes the figure readable: it turns "the number moved" into "the number moved toward or away from real inhibitors". **Source:** the per-iteration `*_comparison_profile.csv` files from the chemical analysis notebooks.

Figure 6 — scaffold and functional-group evolution

(a) Stacked bar of the top 8 Murcko scaffolds' share of $\mathcal{K}_t$ per iteration, with structures drawn as insets for the 3 most frequent. (b) Functional-group frequency $f_{g,t}$ trajectories, one line per group, with the actives' frequency as a dashed horizontal reference per group.

Figure 7 — descriptor-space PCA

Actives (grey, fitted space) with $\mathcal{K}_t$ overlaid per iteration in a sequential colourmap (light $\rightarrow$ dark with increasing t). One panel per target. If the loop is working, later iterations should visibly migrate into the grey cloud. If they do not, that is also worth showing. **Critical:** fit the PCA on actives only, transform both. Do not refit per iteration or panels stop being comparable.

3.6 Rediscovery hit catalogue

Write 1–2 paragraphs + fill Table 9

For each molecule with $\text{sim}_{\max} \geq \theta_{\text{hit}}$: report the iteration it appeared, its similarity, its nearest known active with that active's ChEMBL ID and measured pIC_{50} , and Boltz-2's predicted pIC_{50} for the generated molecule. If there are no formal hits, retitle this subsection "Closest generated–active pairs", report the top 5–10 by $\text{sim}_{\max}$, and say plainly that no molecule cleared 0.85. That is an honest and perfectly reportable outcome.

Table 9: Rediscovery hits and closest generated–active pairs.

Target	Iter	$\text{sim}_{\max}$	Nearest active (ChEMBL ID)	Measured pIC_{50}	Predicted pIC_{50}	PAINS

Figure 8 — hit catalogue grid

Side-by-side 2D structures: generated molecule (left) next to its nearest known active (right), one row per hit, most similar first, annotated with $\text{sim}_{\max}$ and both potency values. **Source:** `Draw.MolsToGridImage` in the Part 2 analysis notebook already produces exactly this (`*_closest_pairs.png`); increase `subImgSize` and save as SVG or PDF for the paper.

3.7 Cross-target comparison

Write 2 paragraphs

This subsection is the generalisability claim, so it needs care. Compare across targets: calibration quality (AUC, correlation), convergence rate ($\Delta \bar{T}$ per iteration, not absolute $\bar{T}$), hit iteration, and scaffold behaviour. Explicitly control for the confounds: different $|\mathcal{A}|$ (3,860 vs. 1,402), possibly different τ , and different pocket character. If the two targets behaved similarly, that supports generalisation; if they diverged, identify which of these three confounds is the likeliest explanation and say what experiment would separate them.

[TODO: cross-target comparison]

3.8 Ablations

Optional — include only what you actually ran

Candidate ablations, roughly in order of value per unit of GPU time:

1. **No-gate control:** rank by $-\hat{y}$ alone with no p_{bind} filter. Directly tests whether the gate matters. Cheapest and most informative.
2. **Alternative scoring** (Options A and C from Section 2.6).
3. **Gate sensitivity:** rerun with $\tau \pm 0.1$.
4. **Corpus size:** $K = 200$ vs. 500 — probes the exploitation/diversity trade-off.

5. **Random-corpus control:** fine-tune on K *randomly chosen* passing molecules instead of the top- K . If $\bar{T}_t$ rises anyway, your improvement is coming from fine-tuning on anything, not from the scoring signal. This is the single most convincing control in the whole paper — run it if you run only one.

Delete this subsection entirely if you ran none.

[TODO: ablations, or delete this subsection]

4 Discussion

Optional section — keep if the venue expects it

Four short blocks:

1. **What the results mean.** One paragraph tying the convergence curves back to the claim in Section 1.7.
2. **Limitations.** Predicted rather than measured affinity; possible overlap between Boltz-2’s training data and the benchmark actives; two kinase targets do not test non-kinase generalisation; no synthesis or assay; single random seed if that is the case.
3. **Comparison to prior work.** Where your numbers sit relative to the closed-loop papers reviewed in Section 1.4.
4. **Future work.** A non-kinase target; multi-objective scoring with SA/QED terms in the objective rather than only as monitoring; an experimental validation arm.

Do not use the limitations paragraph to pre-empt criticism you could instead fix.

[TODO: discussion, or delete]

Author contributions

CRedit taxonomy — fill in before submission

Use the standard categories: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Supervision, Funding acquisition. Every author needs at least one. Agree on this early; it is far more awkward to negotiate the week before submission.

[TODO: author contributions]

Acknowledgements

Do not forget the ACCESS citation

Computational work used the Anvil cluster at Purdue University through NSF ACCESS allocation `bio220114`. ACCESS requires a specific acknowledgement wording and citation — look it up on the ACCESS site and use their exact text.

[TODO: acknowledgements + ACCESS allocation wording]

References

- [1] Marcus Olivecrona, Thomas Blaschke, Ola Engkvist, and Hongming Chen. Molecular de-novo design through deep reinforcement learning. *Journal of Cheminformatics*, 9(1):48, 2017. doi: 10.1186/s13321-017-0235-x.

- [2] Thomas Blaschke, Josep Arús-Pous, Hongming Chen, Christian Margreitter, Christian Tyrchan, Ola Engkvist, Kostas Papadopoulos, and Atanas Patronov. REINVENT 2.0: An AI tool for de novo drug design. *Journal of Chemical Information and Modeling*, 60(12):5918–5922, 2020. doi: 10.1021/acs.jcim.0c00915.
- [3] Hannes H. Loeffler, Jiazhen He, Alessandro Tibo, Jon Paul Janet, Alexey Voronov, Lewis H. Mervin, and Ola Engkvist. Reinvent 4: Modern AI-driven generative molecule design. *Journal of Cheminformatics*, 16(1):20, 2024. doi: 10.1186/s13321-024-00812-5.
- [4] Marwin H. S. Segler, Thierry Kogej, Christian Tyrchan, and Mark P. Waller. Generating focused molecule libraries for drug discovery with recurrent neural networks. *ACS Central Science*, 4(1):120–131, 2018. doi: 10.1021/acscentsci.7b00512.
- [5] John Jumper, Richard Evans, Alexander Pritzel, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021. doi: 10.1038/s41586-021-03819-2.
- [6] Josh Abramson, Jonas Adler, Jack Dunger, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500, 2024. doi: 10.1038/s41586-024-07487-w.
- [7] Jeremy Wohlwend, Gabriele Corso, Saro Passaro, et al. Boltz-1: Democratizing biomolecular interaction modeling. *bioRxiv*, 2024. Preprint — verify for an updated journal version.
- [8] Saro Passaro, Gabriele Corso, Jeremy Wohlwend, et al. Boltz-2: Towards accurate and efficient binding affinity prediction. *bioRxiv*, 2025. Preprint — verify for an updated journal version.
- [9] Barbara Zdrazil, Eloy Felix, Fiona Hunter, et al. The ChEMBL database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1): D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.
- [10] Greg Landrum and RDKit contributors. RDKit: Open-source cheminformatics. <https://www.rdkit.org>, 2026. Record the exact version you used.
- [11] David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.
- [12] Guy W. Bemis and Mark A. Murcko. The properties of known drugs. 1. molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996. doi: 10.1021/jm9602928.
- [13] Christopher A. Lipinski, Franco Lombardo, Beryl W. Dominy, and Paul J. Feeney. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Advanced Drug Delivery Reviews*, 23(1–3):3–25, 1997. doi: 10.1016/S0169-409X(96)00423-1.
- [14] G. Richard Bickerton, Gaia V. Paolini, Jérémy Besnard, Sorel Muresan, and Andrew L. Hopkins. Quantifying the chemical beauty of drugs. *Nature Chemistry*, 4(2):90–98, 2012. doi: 10.1038/nchem.1243.
- [15] Peter Ertl and Ansgar Schuffenhauer. Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *Journal of Cheminformatics*, 1(1):8, 2009. doi: 10.1186/1758-2946-1-8.
- [16] Jonathan B. Baell and Georgina A. Holloway. New substructure filters for removal of pan assay interference compounds (PAINS) from screening libraries and for their exclusion in bioassays. *Journal of Medicinal Chemistry*, 53(7):2719–2740, 2010. doi: 10.1021/jm901137j.